640, 641

Air University
End Semester Examinations: Fall 2025

AIR UNIVERSITY

16-12-2025

Student ID: ________

Subjective Part
(To be solved on Answer Books only)

Subject: Calculus and Analytical Geometry
Class: BSCYS-III
Section(s): A, B (Morning Session)
Course Code: MA-110

Time Allowed: 180 Minutes
Max Marks: 100
Date: 17-JAN-2026
Time: 09:00-12:00
FM's Name: Mr. Umair Habib
FM's Signature:

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.
- Tables/calculators are allowed.

Q1 (CLO-1) **Marks (5+5+5)**

a) **Find** the domain and range of the given function. $f(x) = \frac{2x-1}{x+4}$

b) Sketch the graph of the following function by using Translation and reflection as appropriate. $f(x) = 6 - x - x^2$ (**NOTE:** Sketch all the versions before the final graph).

c) **State** conditions under which two functions, f & g, will be the inverses of one another, and give at least 3 examples of such functions.

Q2 (CLO-2) **Marks (5+5+5+5)**

a) **Discuss** the continuity of the function $f(x) = \frac{1}{\sqrt{x}}$ on the interval $[1,9]$.

b) **Interpret** the limit. $\lim_{x\to+\infty} \sqrt{x^6 + 5x^3} - x^3$

c) If $f(x) = \frac{Sin(x)-Sin(1)}{x-1}$. Use the identity $Sin(\alpha) - Sin(\beta) = 2Cos\left(\frac{\alpha+\beta}{2}\right)Sin\left(\frac{\alpha-\beta}{2}\right)$ to **describe** $\lim_{x\to1} f(x)$.

d) Check (**demonstrate**) whether the function $f(x) = |x|$ is differentiable or not at $x = 0$.

Q3 (CLO-3) **Marks (5+8+5+7)**

a) **Calculate** the area between the graph of the function f and the interval $[a, b]$.
$f(x) = \sqrt{1-x^2}$; $[a, b] = [-1,1]$ & $n = 5$

b) Use Integration by Parts to **Solve.** $\int e^x Sin(x)Cos(x)dx$

c) **Solve** the Initial Value Problem $\frac{dy}{dt} = Sec^2t - Sint$; $y\left(\frac{\pi}{4}\right) = 1$

d) Consider x_k^* as the midpoint of each subinterval to **calculate** the area under the curve $y = f(x)$ over the specified interval. $f(x) = 9 - x^2$; $[0,3]$.

Q4 **(CLO-4)** **Marks (10+10)**

a) Find the radius and height of the right circular cylinder of largest volume that can be inscribed in a right circular cone with radius 6 inches and height 10 inches.

b) Find the point on the graph of the curve $y = 4 - x^2$ which is closest to the point $(3,4)$.

Q5 **(CLO-5)** **Marks (7+13)**

a) Suppose $f(x)$ is a differentiable function with derivative

$f'(x) = (x-1)^2(x-2)(x-4)(x-5)^4$. Find all critical values of $f(x)$ and determine whether each corresponds to a relative maximum, a relative minimum, or neither.

b) Sketch a graph of the polynomial and label the coordinates of the intercepts, stationary points, and inflection points. $p(x) = x(x^2-1)^2$